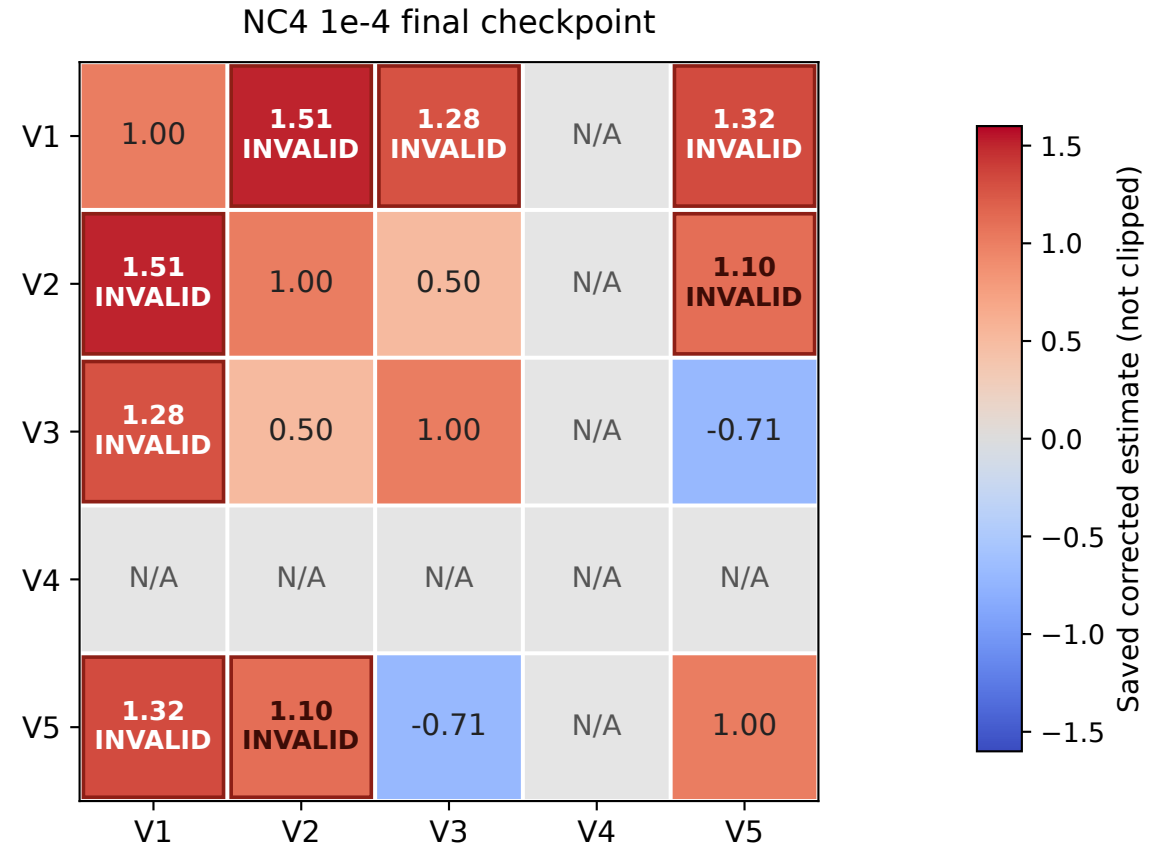
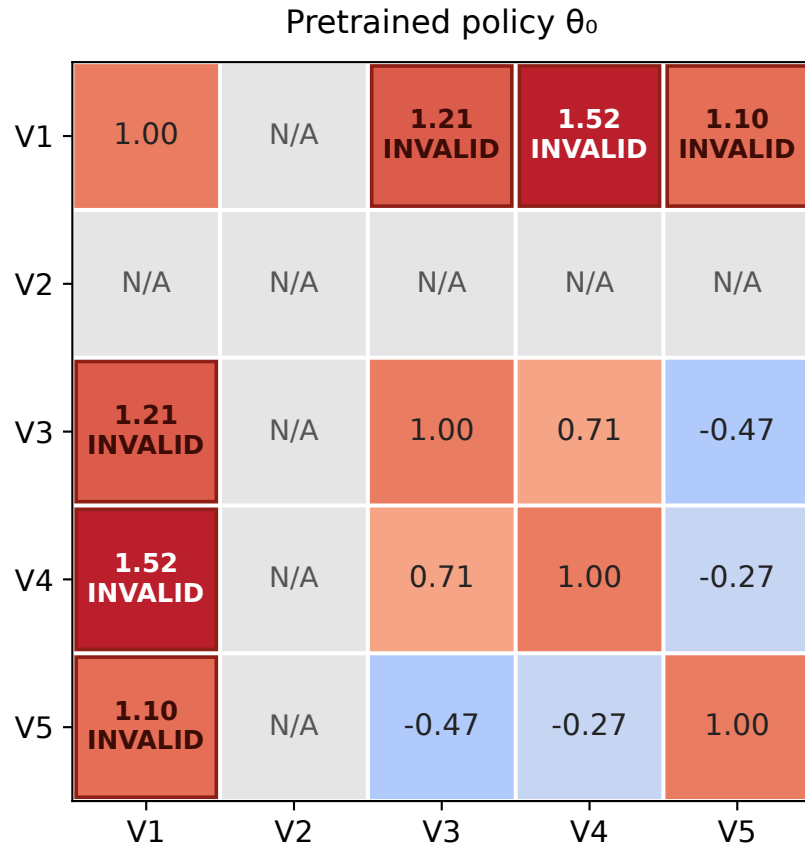


Phase 1: provisional direction-cosine estimates

OUT-OF-RANGE VALUES ARE INVALID COSINES — saved points are shown without clipping; no valid CIs



Red-bordered INVALID cells exceed $[-1, 1]$; they cannot be interpreted as directional alignment.

Gray N/A cells involve a nonpositive corrected squared-signal estimate. These are not measured zero cosines.

V1: scaled GAE (critic); V2: scaled MC – critic; V3: raw MC – LOO. V1-V3 normalize coefficients.

V4: raw MC – LOO with decision discount; V5: raw MC with decision discount. V4-V5 do not normalize.

Between-variant gradient norm scales differ; corrected ratios, including in-range entries, remain provisional.